

1.

1 2 3 4 5

$s = \{c, a, b, o, w\} \rightarrow$  initial dict encoding.

$c = 1$

$cw = 11$

$a = 2$

$wab = 12$

$b = 3$

$bba = 13$

$o = 4$

$acw = 14$

$w = 5$

$wo = 15$

$wa = 6$

$oo = 18$

$ab = 7$

$oc = 17$

$bb = 8$

$cwo = 18$

$ba = 9$

$ac = 10$

$wabbaacwabbacwwoacwwo$   
5 2 3 3 1 6 8 10 5 4 4 1 16

Decoding

$1 = c$

$10 = ac$

$2 = a$

$11 = cw$

$3 = b$

$12 = wab$

$4 = o$

$13 = bba$

$5 = w$

$14 = acw$

$6 = wa$

$15 = wo$

$7 = ab$

$16 = oo$

$8 = bb$

$17 = oc$

$9 = ba$

$18 = cwo$

|                          |   |   |   |   |    |    |    |    |    |    |    |    |    |    |
|--------------------------|---|---|---|---|----|----|----|----|----|----|----|----|----|----|
| <input type="checkbox"/> | s | 2 | 3 | 3 | 2  | 1  | 6  | 8  | 10 | 5  | 9  | 9  | 11 | 16 |
| <input type="checkbox"/> | w | a | b | b | a  | c  | wa | bb | ac | w  | o  | o  | ow | oo |
| <input type="checkbox"/> |   |   |   |   |    |    |    |    |    |    |    |    |    |    |
| <input type="checkbox"/> | 0 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |    |

2.

 $S = \{a, r, s, y\}$  initial dict

encoding

a = 1

s = 11

r = 2

sa = 12

s = 3

ar = 13

r = 4

rr = 14

s = 5

ray = 15

ab = 6

y s y = 16

sr = 7

ya = 17

ra = 8

as = 18

ay = 9

ss = 19

ys = 10

sar = 20

 $a\$ray \quad \$sarray \quad \$yassaray$ 

1 5 2 1 4 5 3 1 2 8 10 4 1 3 1 2 1 5

decoding

1 = a

6 = a\$

11 = \$s

16 = ysy

2 = r

7 = \$r

12 = \$a

17 = ya

3 = s

8 = ra

13 = ar

18 = as

4 = y

5 = ay

14 = rr

19 = ss

5 = s

10 = ys

15 = ray

20 = sar

9. array based ray YB. Y-a, S-a ray

6 7 8 9 10 11 12 13 14 15 16 17 18 19 20.

3.

$$a = 1$$
$$+b = 14$$
$$b = 2$$
$$b_1 = 15$$
$$h = 3$$
$$1.5h = 1.7$$
$$1 = 4$$
$$bhl = 17$$
$$S = 5$$
$$15bh = 10.$$
 $t = 6$ 
$$h_{at} = 19$$
$$h = 7$$
$$x_{b1} = 20$$
$$h_1 = 8$$

1+ - 21

$$\dot{y} = g$$
$$7615 = 22$$
$$S_b = 10$$
$$sh\ h = 23$$
$$bh = 11$$
$$n_{15} = 2^4$$
$$h_a = 12$$
$$gha = 25$$
$$at = 13$$

t h i s b h a t b i s b h i s b h a t b i i t b i s b h i s b h a t

7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25